ECON 0100 | Vignette E1 | Costs

Solutions

Answers and work are shown in the red blocks.

Due in Recitation. Vignettes are a certificate of the work done together in Recitation.

As an economics student at Hogwarts, the Ministry of Magic has tasked you with analyzing the impact of policy proposals on three firms in the cauldron industry, given their distinct cost structures:

The Cauldron Co. is a large, established firm producing high-quality cauldrons with traditional methods. They have high fixed costs but low, constant marginal costs due to economies of scale.

The Witching Hour is a smaller firm specializing in novelty cauldrons. They have moderate fixed costs but increasing marginal costs due to specialized production.

Nimbus Cauldron Works is a new entrant producing low-cost, mass-produced cauldrons made to order by a contract foundry. They have no meaningful fixed costs, and low constant marginal costs.

Q1 | The Witching Hour's Workshop

The table shows how many cauldrons The Witching Hour makes in a day with each number of workers. Each worker is paid 60 coins a day, and the workshop pays 120 coins a day in rent.

L	Q	Marginal Product	Marginal Cost	Variable Cost	Total Cost	ATC
0	0	—	—	0	120	—
1	15	15	4	60	180	12
2	24	9	6.67	120	240	10
3	30	6	10	180	300	10
4	32	2	30	240	360	11.25

Fill in the marginal product of each worker. Then fill in the marginal cost, variable cost, total cost, and average total cost at each number of workers.

> Marginal product is the extra cauldrons from one more worker. Marginal cost is what each of those cauldrons adds to total cost: the wage divided by the marginal product, $60/15 = 4$, $60/9 = 6.67$, $60/6 = 10$, $60/2 = 30$. Variable cost is the wage bill, $60 \times L$. Total cost adds the rent of 120. Average total cost is total cost divided by cauldrons.

a) What happens to the marginal product as workers are added?

It falls: 15, 9, 6, 2. Each added worker adds fewer cauldrons than the one before.

b) Why does The Witching Hour have increasing marginal costs?

Every worker costs the same 60 coins, but later workers add fewer cauldrons, so each extra cauldron costs more to make. Diminishing marginal product is the reason marginal cost rises.

Q2 | Cost Curves and Policies

Graph all three firms' average total cost and marginal cost curves. Then show how each firm's curves shift with the policy listed.

The original curves (dashed in the graphs).

The Cauldron Co. Marginal cost is constant, so MC is a flat line. The high fixed cost is spread over more cauldrons as output grows, so $ATC = MC + FC/q$ starts high and falls toward MC without ever reaching it.

The Witching Hour. Marginal cost rises with output, so MC slopes up. At low output the fixed cost dominates and ATC falls; once MC rises above ATC it pulls the average up. ATC is U-shaped and MC crosses it at its lowest point.

Nimbus Cauldron Works. With no fixed cost and a constant marginal cost, every cauldron costs the same to make, so the average equals the marginal: $ATC = MC$, one flat line.

The shifts (solid).

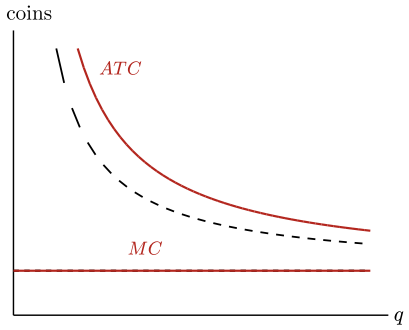
The Cauldron Co. A lump sum tax is the same amount however much the firm makes, so it acts like a fixed cost. Marginal cost does not move. Average total cost rises by the tax divided by output, so it rises most at low output and the gap closes as output grows.

The Witching Hour. A per unit tax is paid on every cauldron, so every cauldron costs that much more to make. Marginal cost and average total cost both shift up by the tax at every output, and the lowest point of ATC stays at the same output.

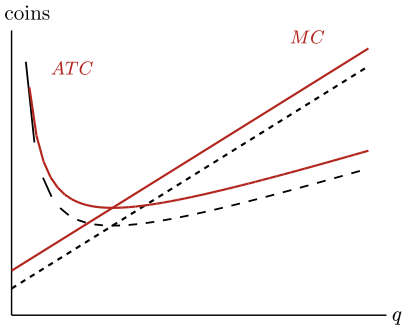
Nimbus Cauldron Works. A per unit subsidy pays the firm on every cauldron, so every cauldron costs that much less to make. Its single flat line, $MC = ATC$, shifts down by the subsidy.

(Drawn with a lump sum tax of 300 coins and a per unit tax and subsidy of 4 coins.)

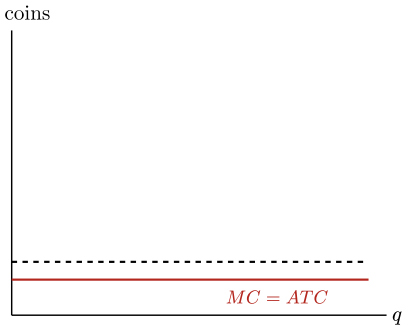
The Cauldron Co.
(lump sum tax)



The Witching Hour
(per unit tax)



Nimbus Cauldron Works
(per unit subsidy)



ECON 0100 | Vignette E2 | Competition

Q1 | A Higher Marginal Cost

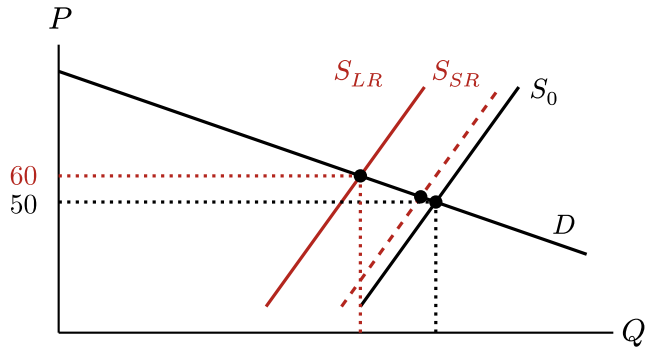
The Witching Hour has a marginal cost of $MC = 10 + 2q$. Many identical workshops sell identical cauldrons, so each takes the market price as given.

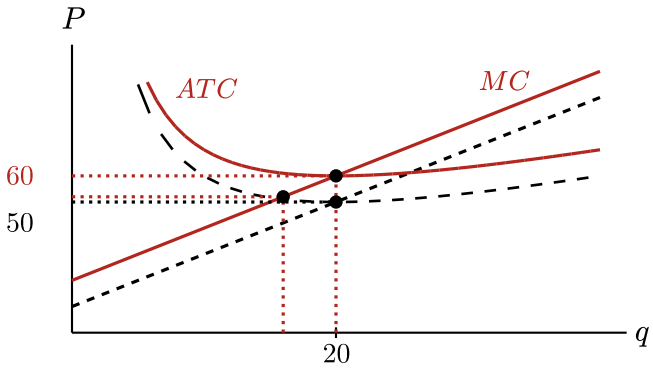
a) At $P = 50$ what quantity maximizes the firm's profit?

> Profit is largest where $P = MC$: $50 = 10 + 2q$, so $q = 20$.

b) Calculate the firm's maximal profit if $ATC = 50$ at the profit-maximizing quantity.

> $(P - ATC) \times q = (50 - 50) \times 20 = 0$. Zero economic profit: the market is in long-run equilibrium. The graphs show the market (left) and one firm (right), before the tax in black and after it in red.





Now suppose the Ministry taxes every cauldron 10 coins, so marginal cost rises: $MC = 20 + 2q$. Start with identical firms earning zero economic profit in long-run equilibrium.

c) What happens to price, each firm's output, and profit in the short run after the new cost?

> Every workshop's MC and ATC shift up by 10, so market supply shifts up. The price rises, but by less than 10 while the same workshops are all still selling. At the new price each workshop makes fewer cauldrons, and the price is below its new ATC, so it makes a loss.

d) In the long run the price rises to 60. What quantity maximizes each remaining firm's profit?

> $P = MC$ with the new marginal cost: $60 = 20 + 2q$, so $q = 20$, the same quantity as before the tax.

e) What happens in the long run to price, each the firm's output, and market output?

> Losses make workshops leave. Market supply shifts in and the price rises until the remaining workshops earn zero economic profit again, at $P = 60$: the price has risen by the full 10. Each remaining workshop is back at its original output of 20. Market output is lower, because buyers want less at 60 than at 50, so there are fewer workshops.